Predicting Term Deposit Subscription

A Machine Learning Approach for Bank Marketing Decision-Making

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- Dataset: Bank Marketing Dataset (UCI)

EXECUTIVE SUMMARY

Predictive targeting significantly improves campaign efficiency by focusing on high-probability customers

- Current campaigns have low conversion (~11%) → inefficient resource use
- Customer behaviour varies strongly by engagement and contact strategy
- Model ranks customers by likelihood of acceptance
- Targeting top segments captures a large share of conversions with fewer calls

BUSINESS PROBLEM

Mass telemarketing drives low conversion while increasing cost and customer fatigue

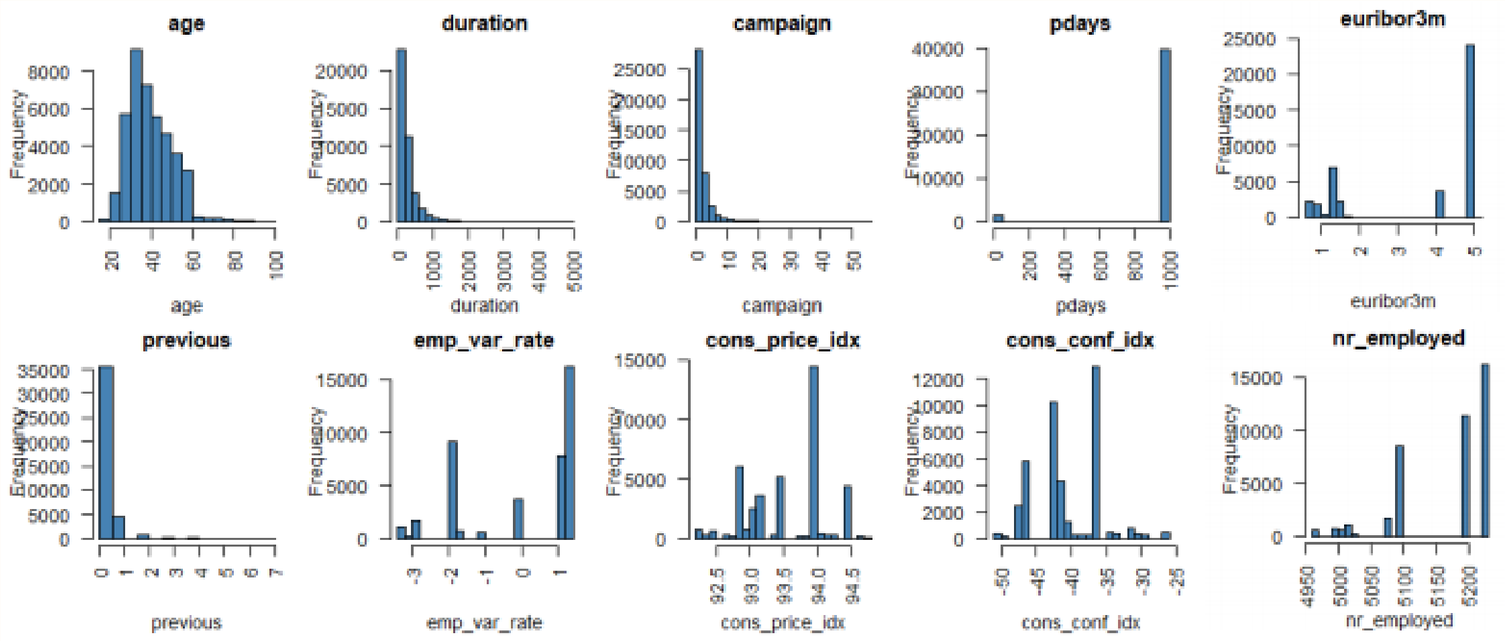
- Low response rate (~11%) limits campaign ROI
- High call volume increases operational cost
- Repeated contact reduces customer responsiveness

INSIGHT 1 (ENGAGEMENT)

Call duration shows strong separation between responders and non-responders

- Longer interactions → higher conversion
- Clear gap between responders and non-responders
- Engagement drives response behaviour

Figure 1: Distribution check of selected variables



Note: Call duration is used for behavioural insight, not for pre-call targeting

INSIGHT 2 (CAMPAIGN FATIGUE)

Excessive contact frequency reduces customer response probability

- Higher campaign frequency → lower response
- Repeated outreach leads to fatigue
- Suggest limiting contact attempts

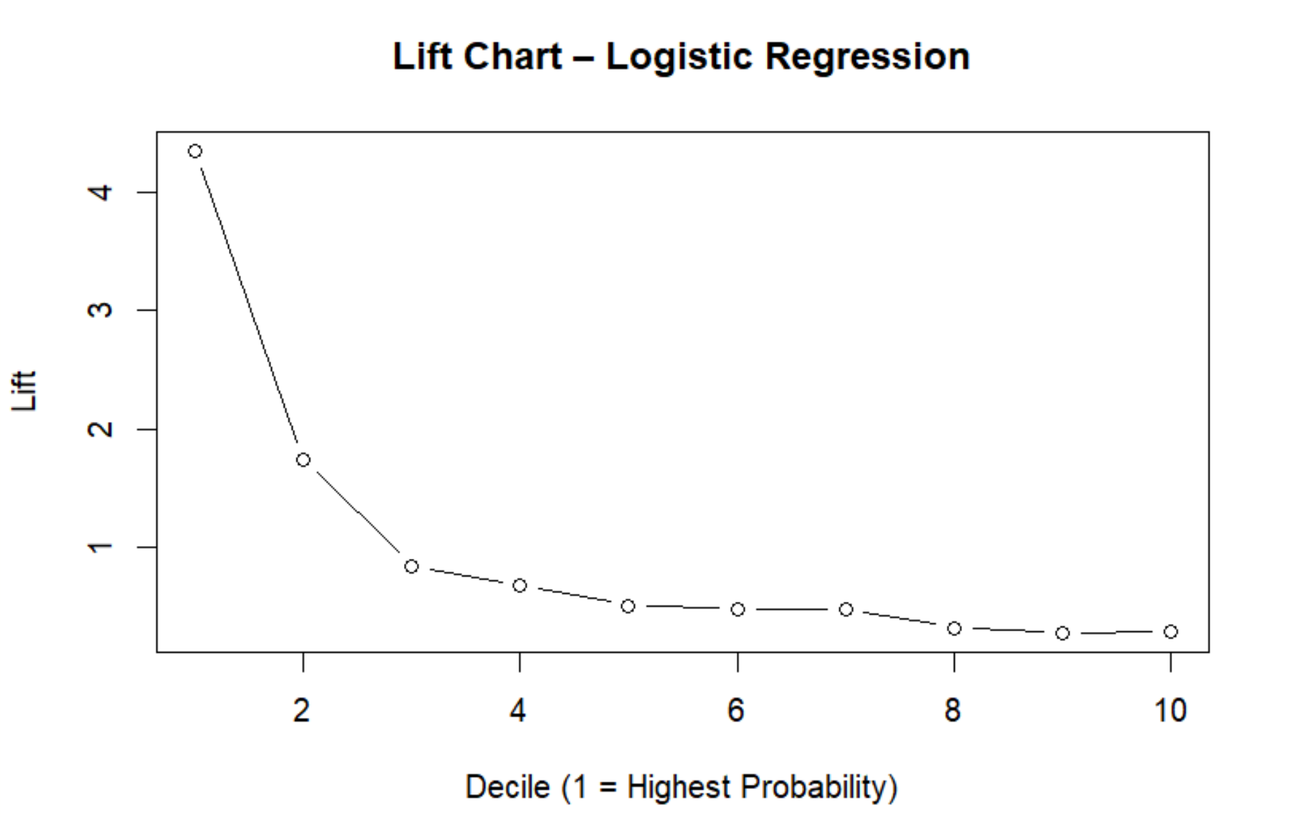
INSIGHT 3 (PRIOR ENGAGEMENT)

Prior customer engagement strongly increases likelihood of response

- Previous contact → higher acceptance probability
- Behaviour persists across campaigns
- Valuable signal for targeting
- Previous contact increases likelihood of response
- Successful past outcome strongly boosts conversion

KEY INSIGHT (LIFT)

Figure 2: Lift Chart – Logistic Regression



The model effectively concentrates conversions in top-ranked customers

- Top-ranked customers capture a disproportionate share of conversions
- Strong lift vs random targeting
- Enables prioritisation of high-value segments
- Enables efficient targeting decisions

SOLUTION

Customer-level probability scoring enables data-driven targeting decisions

- Each customer receives a probability score
- Enables ranking-based targeting instead of mass outreach
- Flexible under budget constraints

MODEL PERFORMANCE

Figure 3: ROC Curve – Logistic Regression

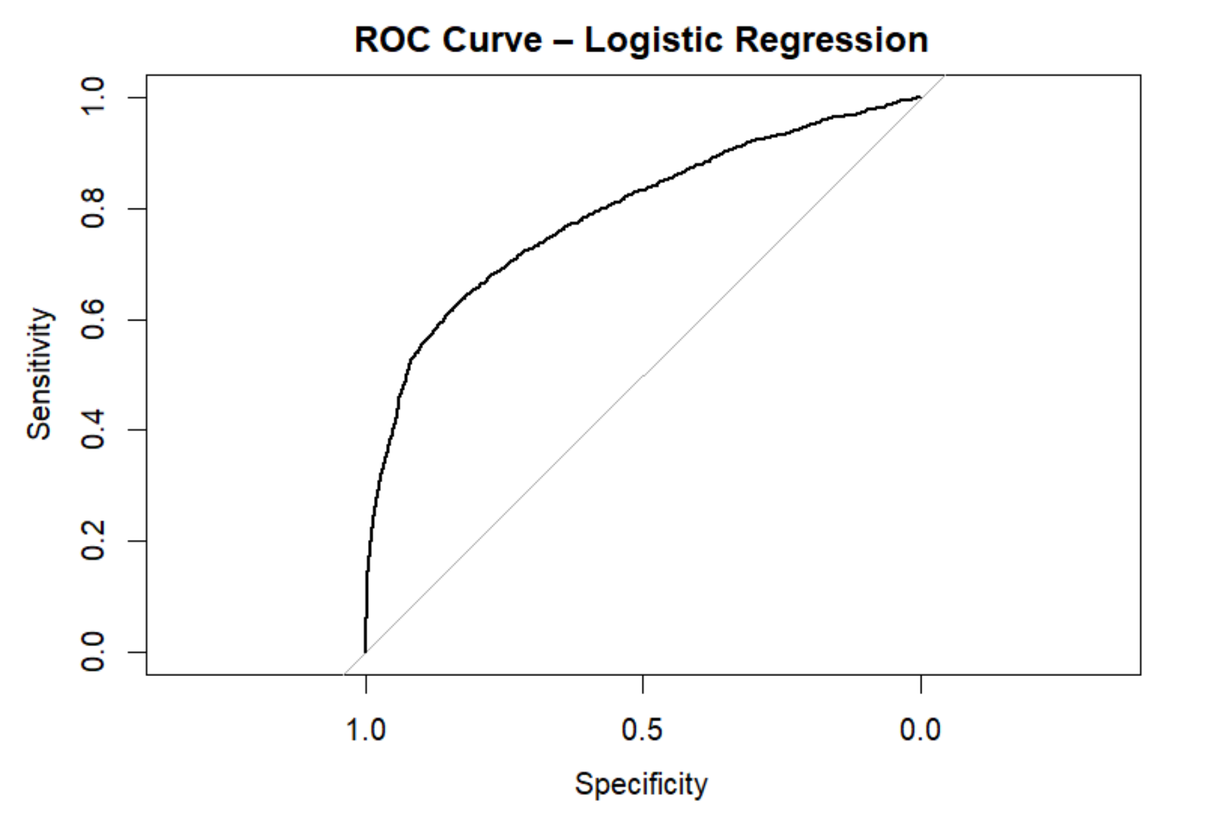
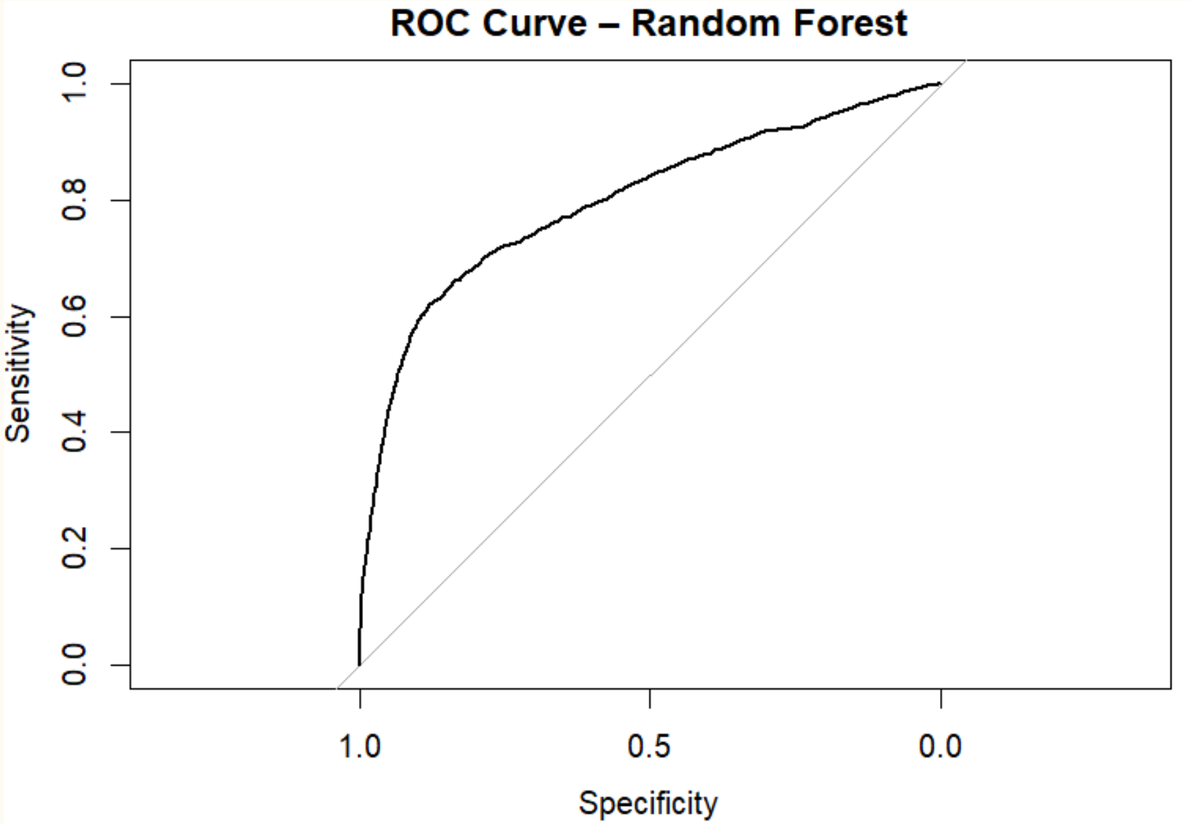


Figure 4: ROC Curve – Random Forest

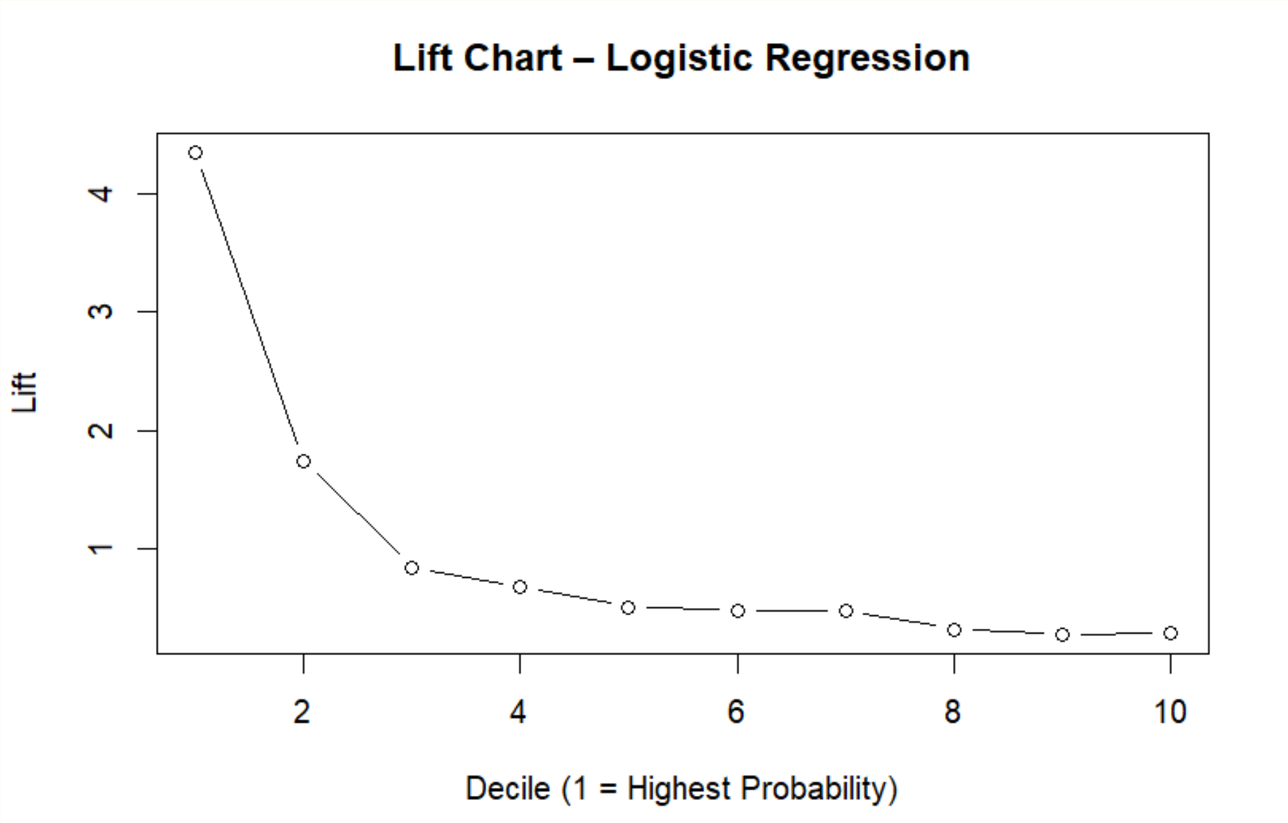


The model delivers strong ranking performance for prioritising customers

- AUC ~0.79 indicates good predictive power
- Random forest shows comparable performance
- Logistic regression preferred for interpretability and deployment
- Model can reliably rank customers for prioritisation

TARGETING STRATEGY

Figure 5: Lift Chart – Logistic Regression



Targeting high-probability customers maximises conversion while reducing cost

- Focus on top 20–40% customers based on probability
- Threshold adjustable based on budget
- Reduces unnecessary outreach

BUSINESS IMPACT

Data-driven targeting improves efficiency and reduces campaign cost

- Higher conversion per contact
- Reduced unnecessary calls
- Improved campaign ROI

CONCLUSION

Predictive targeting enables more effective and scalable marketing decisions

- Customer behaviour is predictable
- Model improves decision quality
- Approach scalable

References

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Thank you!